EWB Courses

Student Project Report

AWS

VPC PEERING

Connect Two VPCs.

Prove Private Connectivity.

A practical project on AWS networking using Amazon VPC, EC2 instances, custom route tables, security groups and a bastion host.

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Abstract

This project focuses on establishing private network connectivity between two Amazon Virtual Private Clouds (VPCs) using VPC Peering in Amazon Web Services (AWS). The architecture consists of two VPCs with non-overlapping IPv4 CIDR blocks, three EC2 instances, custom route tables, an Internet Gateway, and security groups.

VPC A contains a public bastion host and a private application server, while VPC B contains a private application server. The VPC peering connection enables communication between the private workloads using their private IPv4 addresses. Route tables are configured to establish forward and return paths, while security groups restrict inbound traffic to the required sources.

The implementation is validated through connectivity tests before and after peering, bidirectional ICMP testing, private SSH access through the bastion host, and a controlled failure test involving the removal and restoration of an ICMP security group rule.

Project Objectives

The primary objectives of this project are:

1. To create two Amazon VPCs with non-overlapping IPv4 CIDR blocks.
2. To configure public and private subnets with explicit route table associations.
3. To establish a VPC peering connection between the two VPCs.
4. To configure forward and return routes for private communication.
5. To implement security groups that restrict inbound traffic to specified IP addresses.
6. To deploy three EC2 instances, including a public bastion host and two private application servers.
7. To validate connectivity using ICMP and SSH over private IP addresses.
8. To demonstrate controlled connectivity failure and recovery by modifying a security group rule.

Network Architecture and IP Address Plan

5.1 Architecture Description

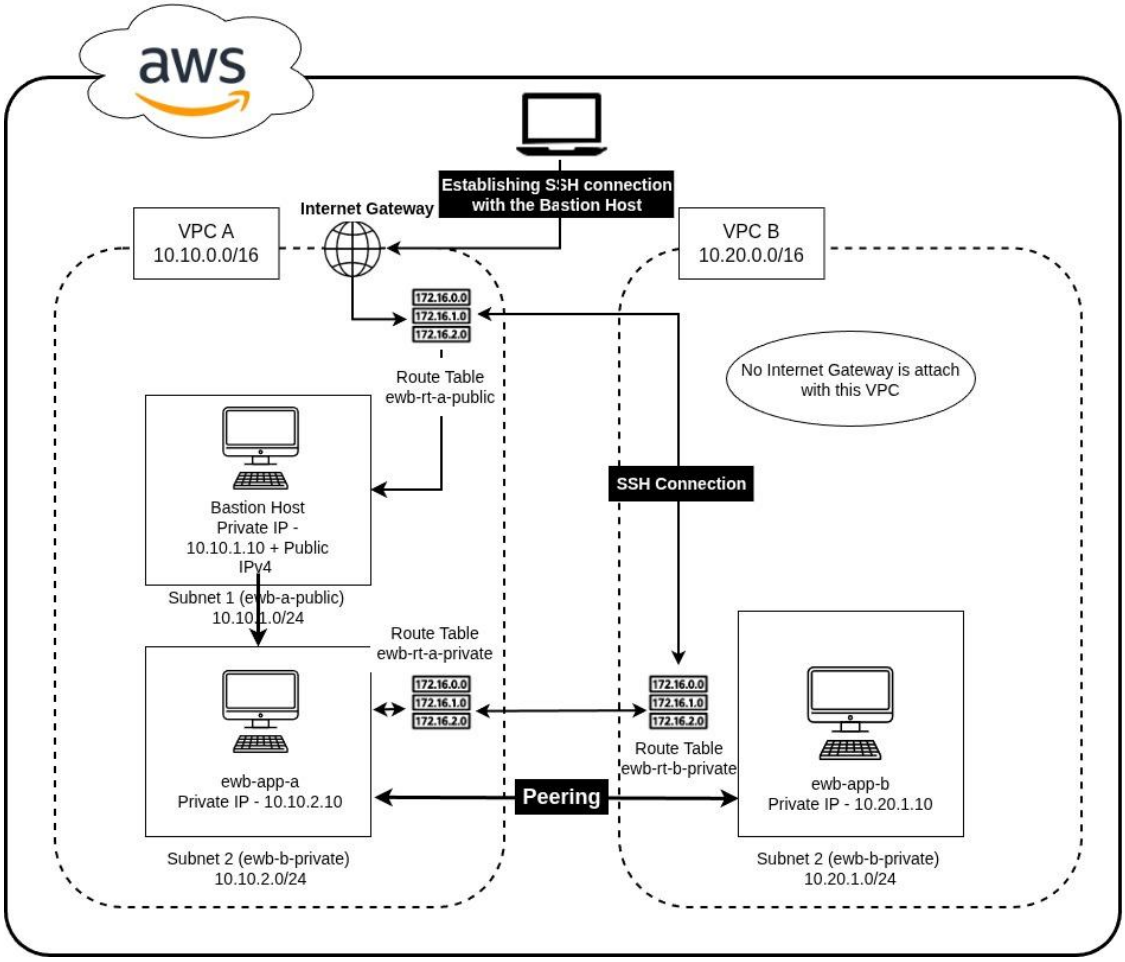


Figure 1 AWS VPC Architecture with Bastion Host and VPC Peering

The project architecture consists of two isolated Amazon VPCs connected through a VPC peering connection. VPC A uses the IPv4 CIDR block 10.10.0.0/16, while VPC B uses 10.20.0.0/16. These address ranges do not overlap, allowing the two VPCs to communicate through peering.

VPC A contains a public subnet for the bastion host and a private subnet for Application Server A. An Internet Gateway provides internet connectivity to the public subnet through its associated route table. VPC B contains a private subnet for Application Server B and does not require an Internet Gateway for this lab.

The bastion host provides the administrative entry point for SSH connections to the private instances. After configuring VPC peering, the application servers communicate using private IPv4 addresses, provided that the necessary routes and security group rules are in place.

5.2 IP Address Plan

Resource	Name	IPv4 CIDR / Address
VPC A	ewb-vpc-a	10.10.0.0/16
Public Subnet A	ewb-a-public	10.10.1.0/24
Private Subnet A	ewb-a-private	10.10.2.0/24
VPC B	ewb-vpc-b	10.20.0.0/16
Private Subnet B	ewb-b-private	10.20.1.0/24
Bastion Host	ewb-bastion	10.10.1.10
Application Server A	ewb-app-a	10.10.2.10
Application Server B	ewb-app-b	10.20.1.10

AWS Network Configuration

VPC and Subnet Configuration

Two Amazon VPCs were configured using non-overlapping IPv4 CIDR blocks. VPC A was assigned 10.10.0.0/16, and VPC B was assigned 10.20.0.0/16.

VPC A contains two subnets: a public subnet with CIDR 10.10.1.0/24 and a private subnet with CIDR 10.10.2.0/24. VPC B contains one private subnet with CIDR 10.20.1.0/24.

The public subnet in VPC A is associated with a route table containing a default route to the Internet Gateway. The private subnets are associated with separate route tables to control communication with their respective VPCs and, after peering, with the remote VPC.

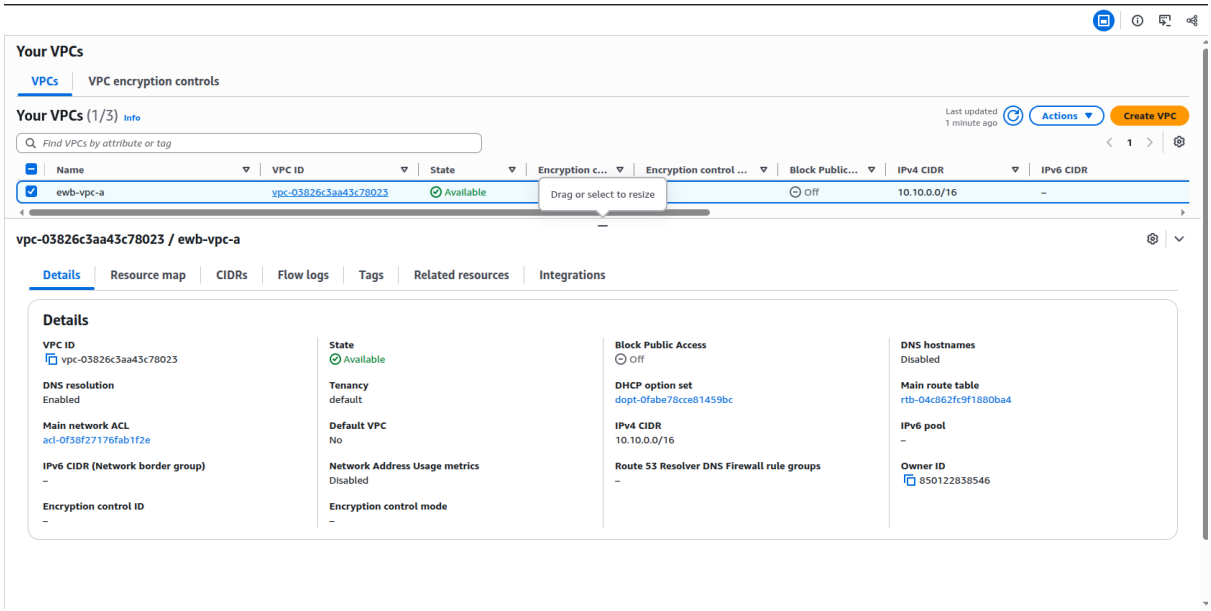


Figure 2 - VPC-A Configuration

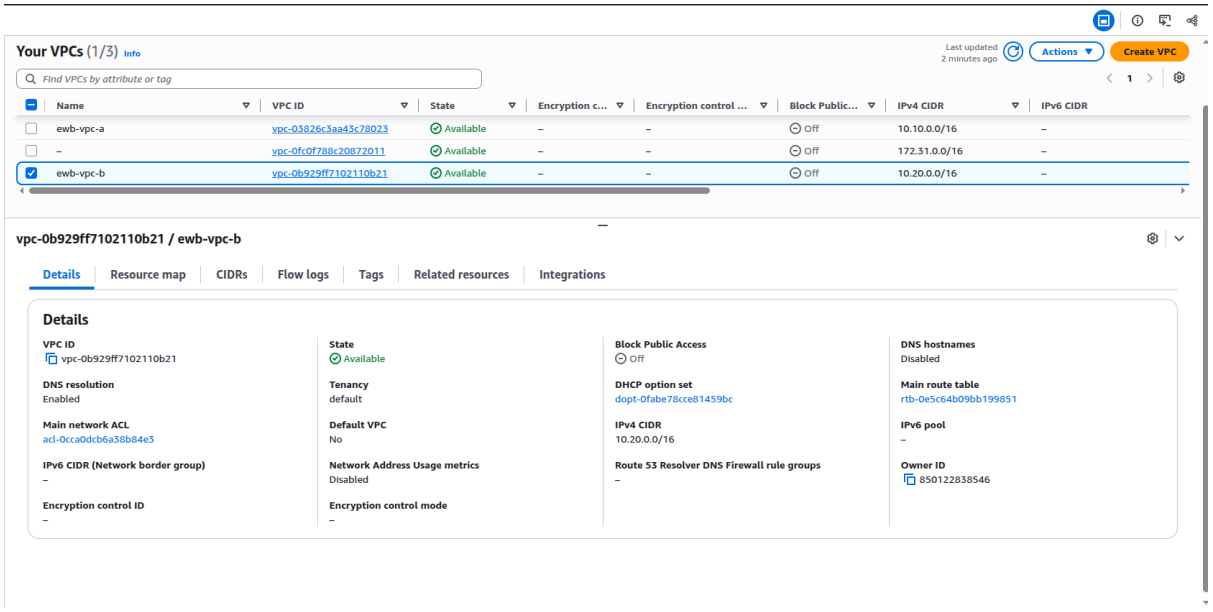


Figure 3 - VPC-B Configuration

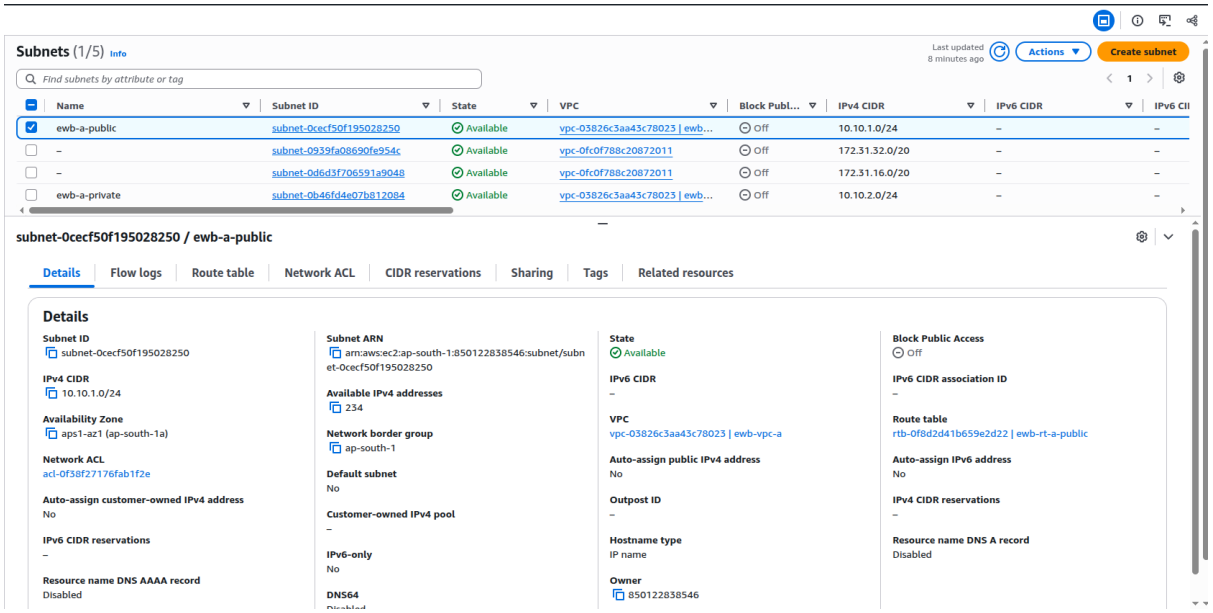


Figure 4 - Public Subnet-A

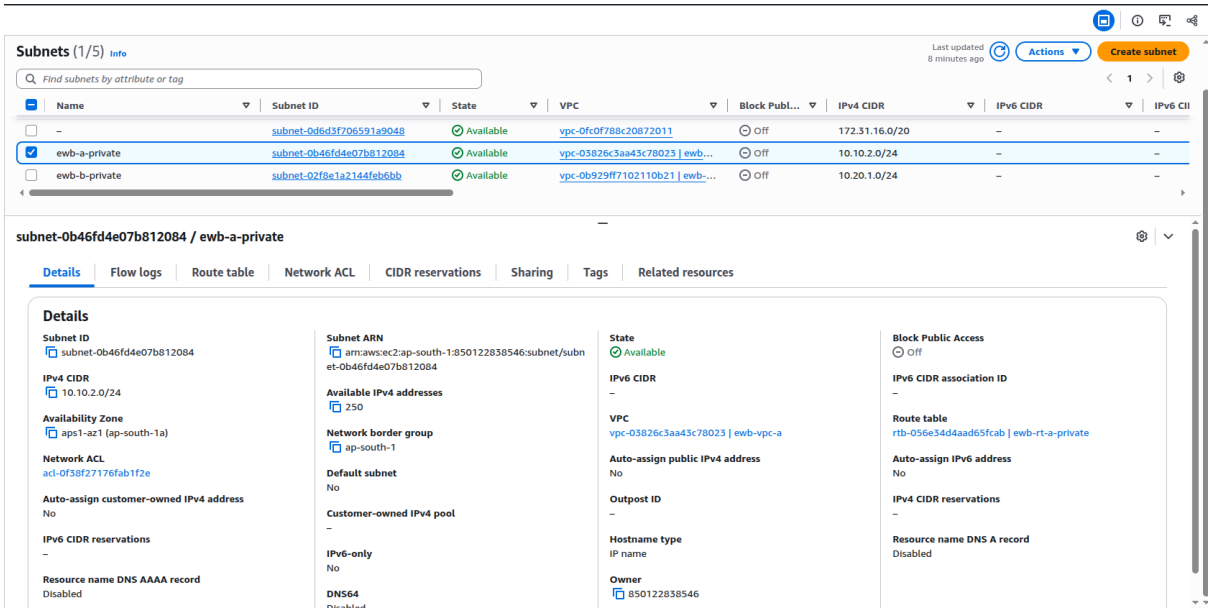


Figure 6 - Private Subnet-A

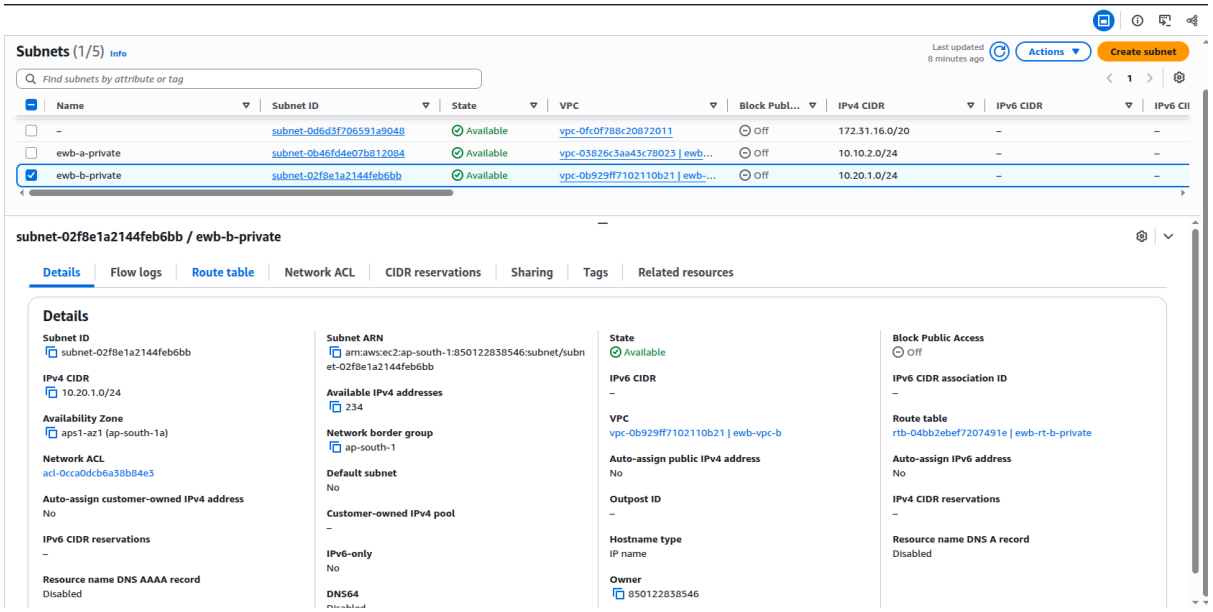


Figure 5 - Private Subnet-B

Internet Gateway

An Internet Gateway named ewb-igw-a was created and attached to VPC A. The public route table was configured with a default route (0.0.0.0/0) targeting this Internet Gateway.

This configuration provides an internet path for the bastion host when it has a public IPv4 address and the required security rules. VPC B does not require an Internet Gateway for private communication through VPC peering.

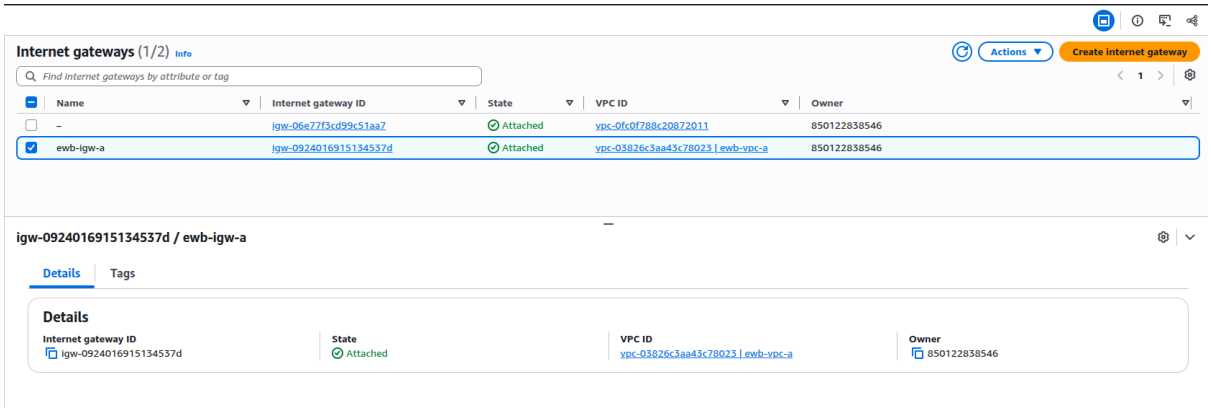


Figure 7 - Internet Gateway

Route Tables: -

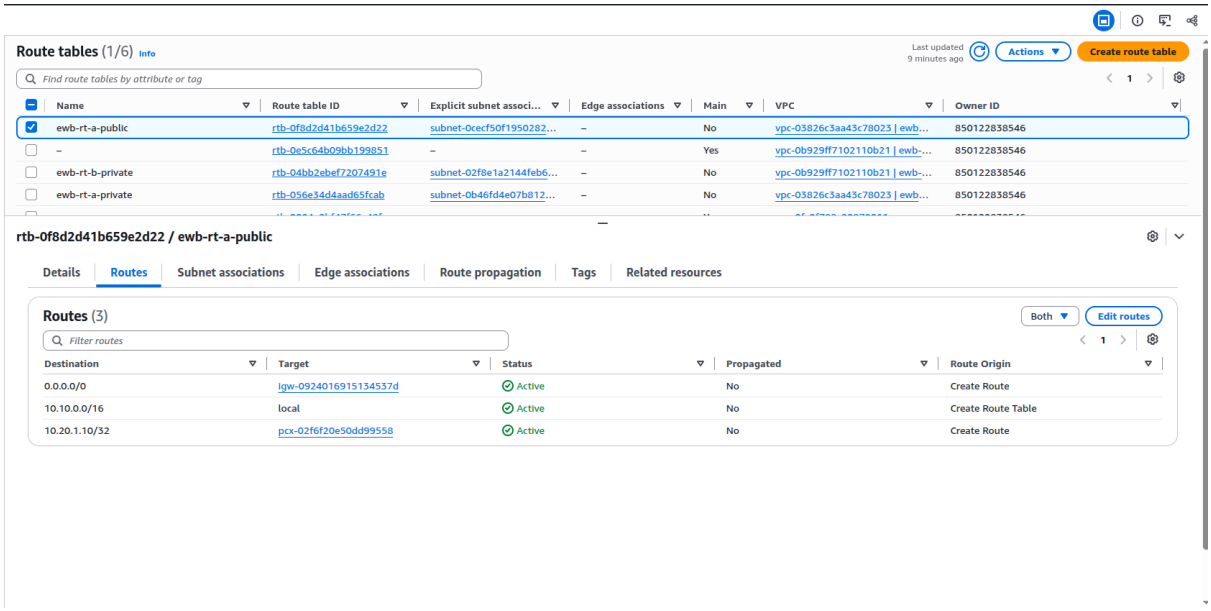


Figure 8 - Route Table for public subnet

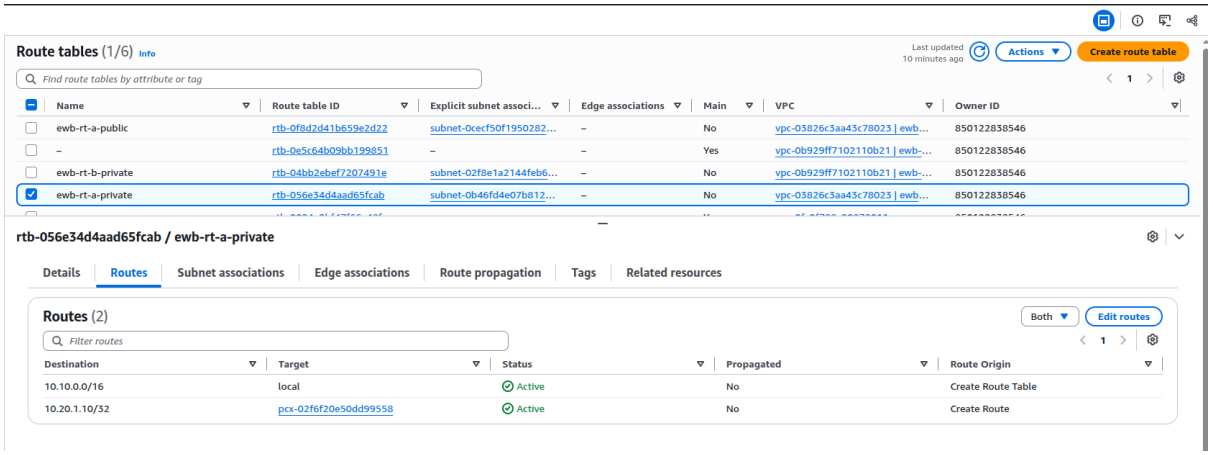


Figure 10 - Private Subnet - A

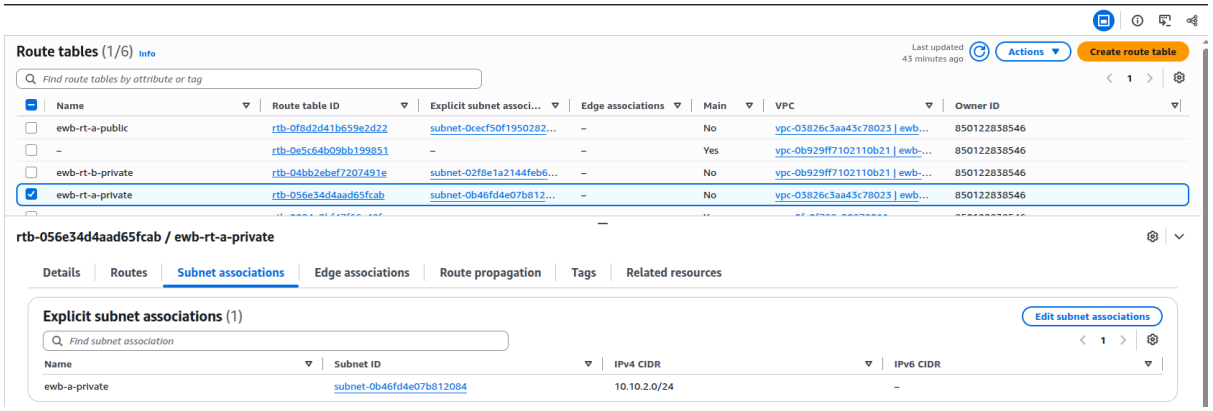


Figure 9 - Route table for private subnet B

Security Group Configuration

Three security groups were configured to control inbound traffic to the bastion host and the two private application servers.

The bastion security group permits SSH traffic on TCP port 22 from the administrator's current public IPv4 address using a /32 source rule. The security group associated with Application Server A permits SSH from the bastion host's private IP address and ICMP Echo Requests from Application Server B.

Similarly, the security group associated with Application Server B permits SSH from the bastion host's private IP address and ICMP Echo Requests from Application Server A. These rules restrict access to the specified sources instead of exposing SSH or ICMP to all IPv4 addresses.

The default outbound allow-all rules and default network ACLs are retained for this lab. Security groups are stateful, so response traffic for permitted connections is allowed automatically.

Security rules table

Security Group	Protocol / Port	Source	Purpose
ewb-sg-bastion	SSH / TCP 22	Your public IPv4 /32	SSH from laptop
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32	SSH through bastion
ewb-sg-app-a	ICMP Echo Request	10.20.1.10/32	Ping from App-B
ewb-sg-app-b	SSH / TCP 22	10.10.1.10/32	SSH through bastion
ewb-sg-app-b	ICMP Echo Request	10.10.2.10/32	Ping from App-A

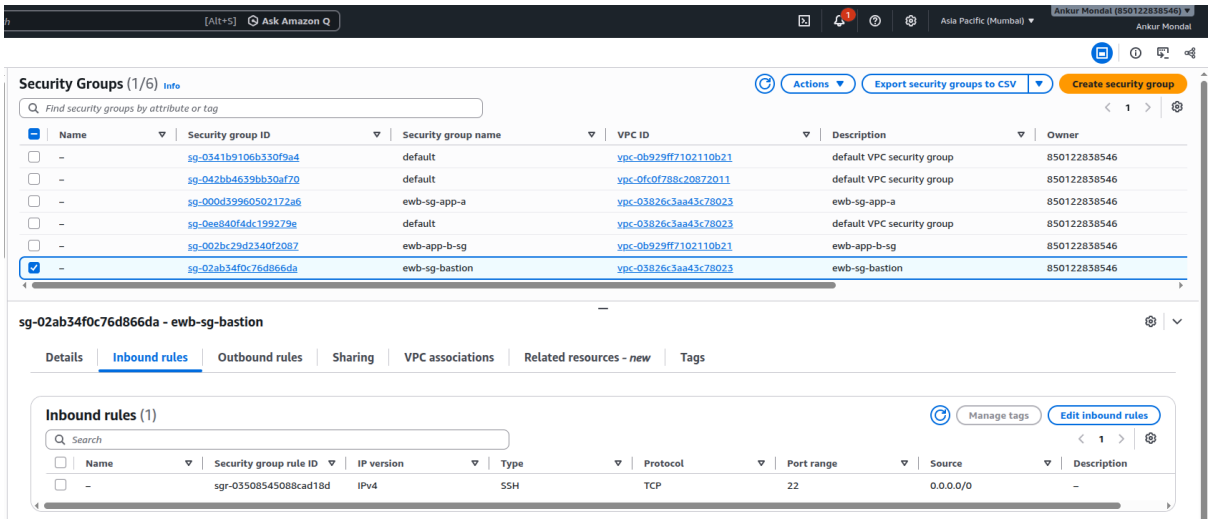


Figure 11 - Bastion security group

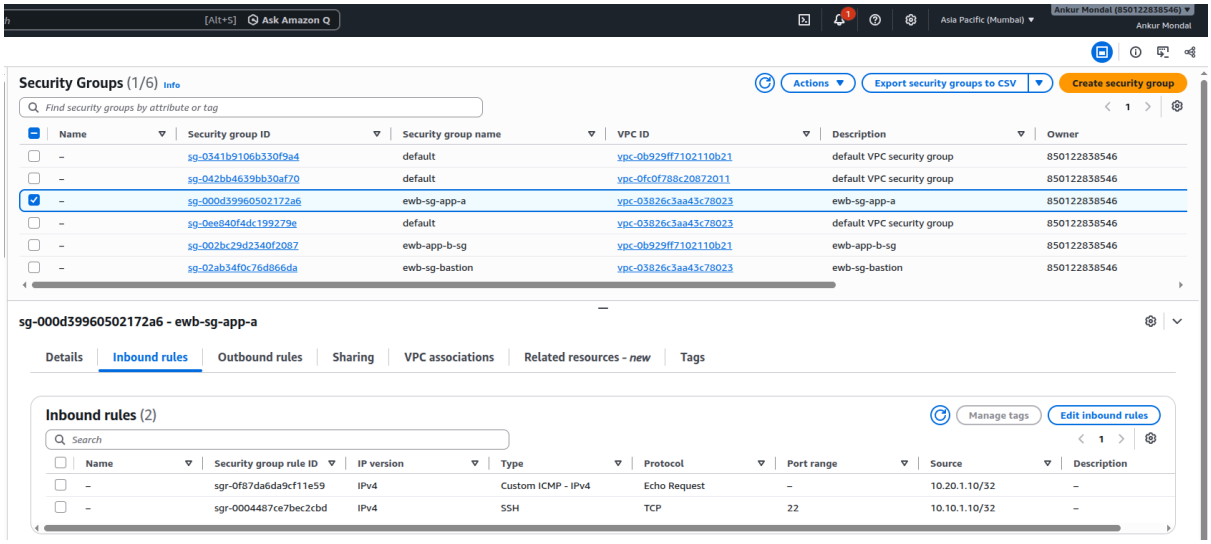


Figure 12 - App-A Security Group

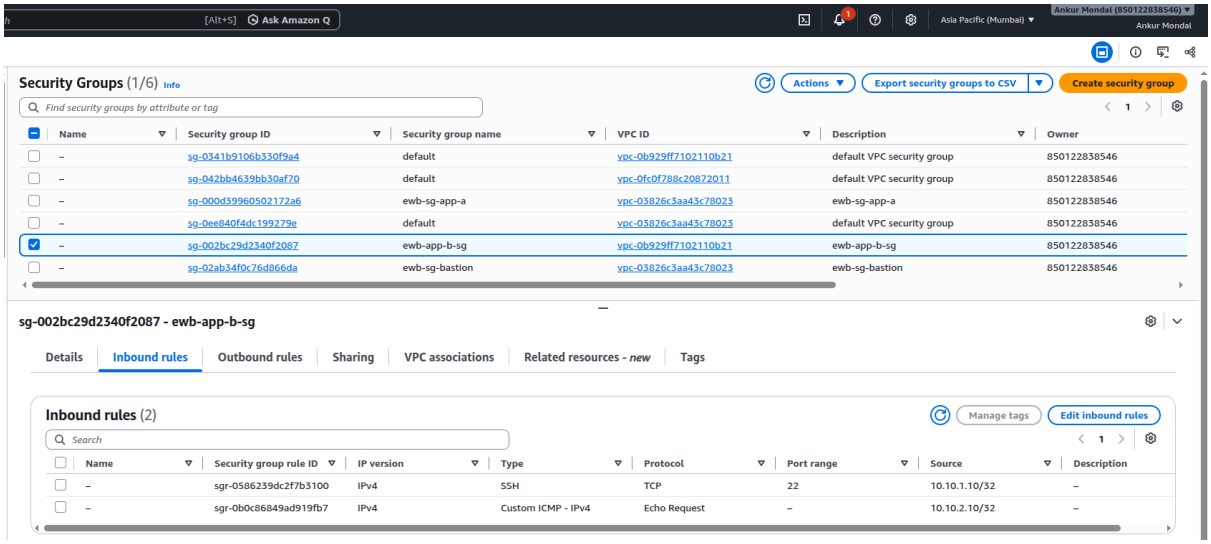


Figure 13 - App-B Security Group

EC2 Instance Configuration

Three Amazon EC2 instances were planned for the project using Amazon Linux 2023 and an instructor-approved small x86 instance type.

The ewb-bastion instance was placed in the public subnet of VPC A and configured with a public IPv4 address. The ewb-app-a instance was placed in the private subnet of VPC A, while ewb-app-b was placed in the private subnet of VPC B.

Both application servers were configured without public IPv4 addresses. The same EC2 key pair was used for SSH authentication, and the bastion host was used as the administrative entry point to the private instances.

Instance	VPC	Subnet	Private IP	Public IP
ewb-bastion	VPC A	Public	10.10.1.10	[Actual public IP]
ewb-app-a	VPC A	Private	10.10.2.10	None
ewb-app-b	VPC B	Private	10.20.1.10	None

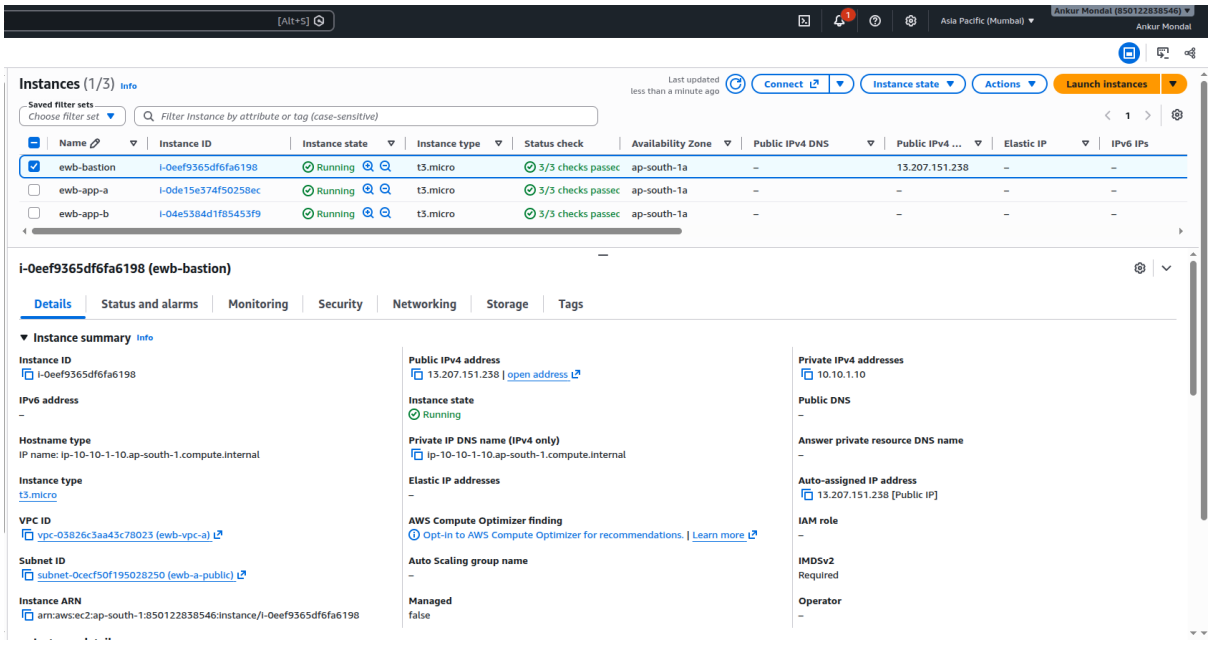


Figure 14 - Bastion Host instance

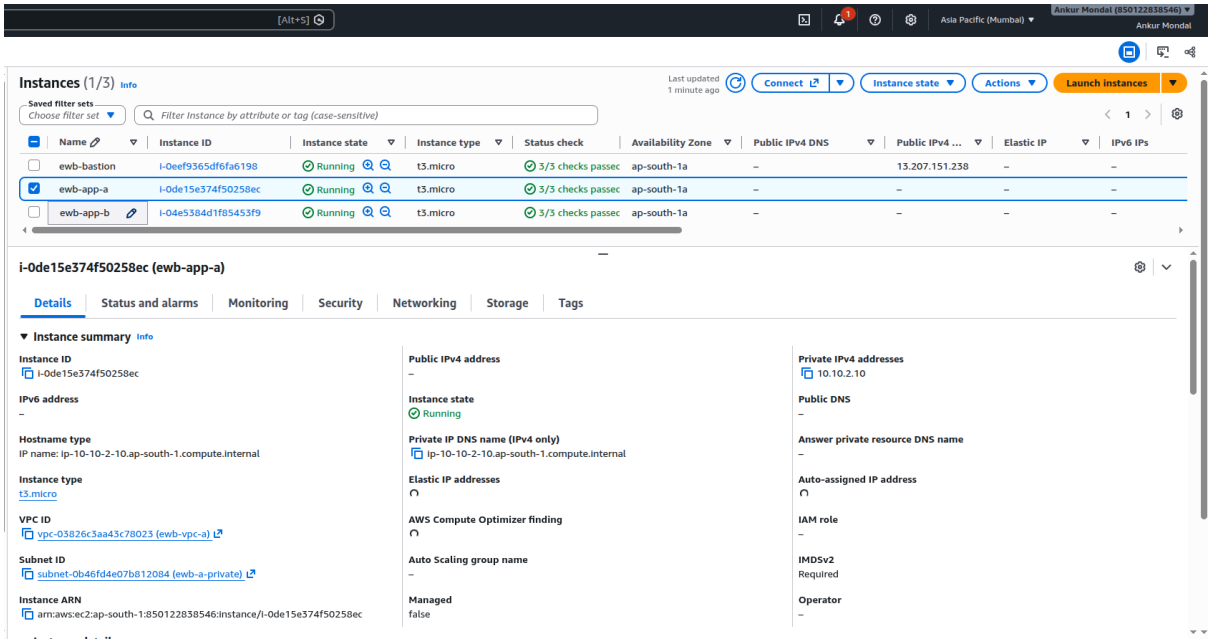


Figure 15 - App-A

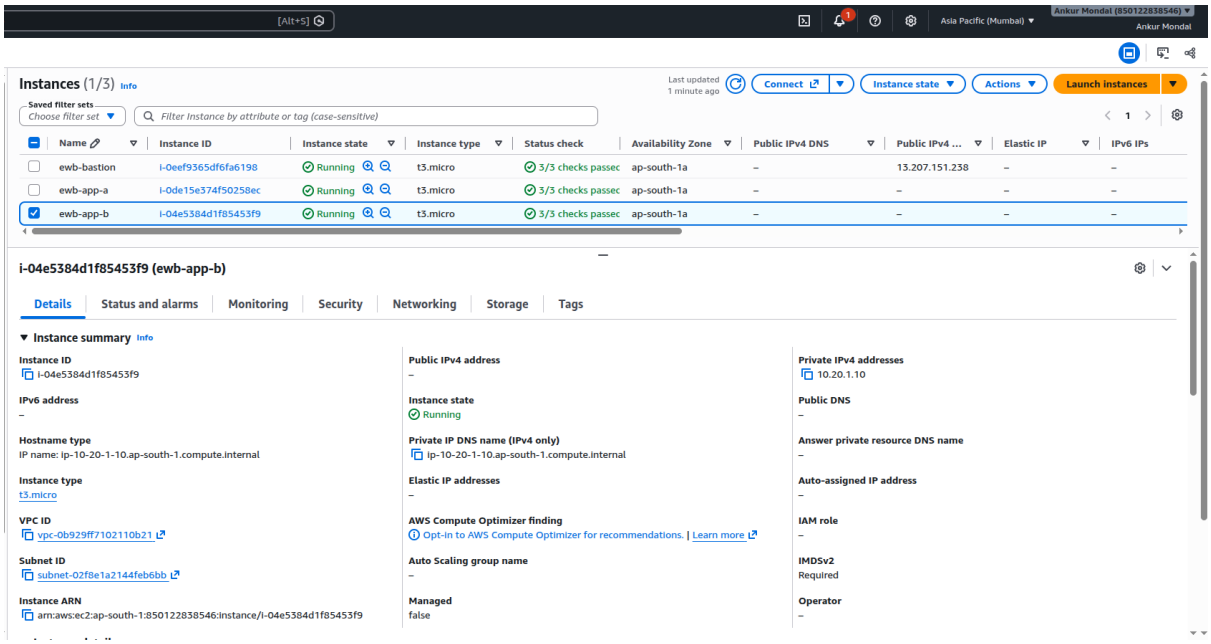


Figure 16 - App-B

VPC Peering Configuration

A VPC peering connection named ewb-a-b-peer was created between VPC A and VPC B within the same AWS account and Region.

VPC A was configured as the requester, and VPC B was configured as the accepter. The peering request was accepted, and the connection status was checked in the AWS VPC console.

The peering connection provides a private network path between the two VPCs. However, communication also requires appropriate routes in both directions and security group rules that permit the intended traffic.

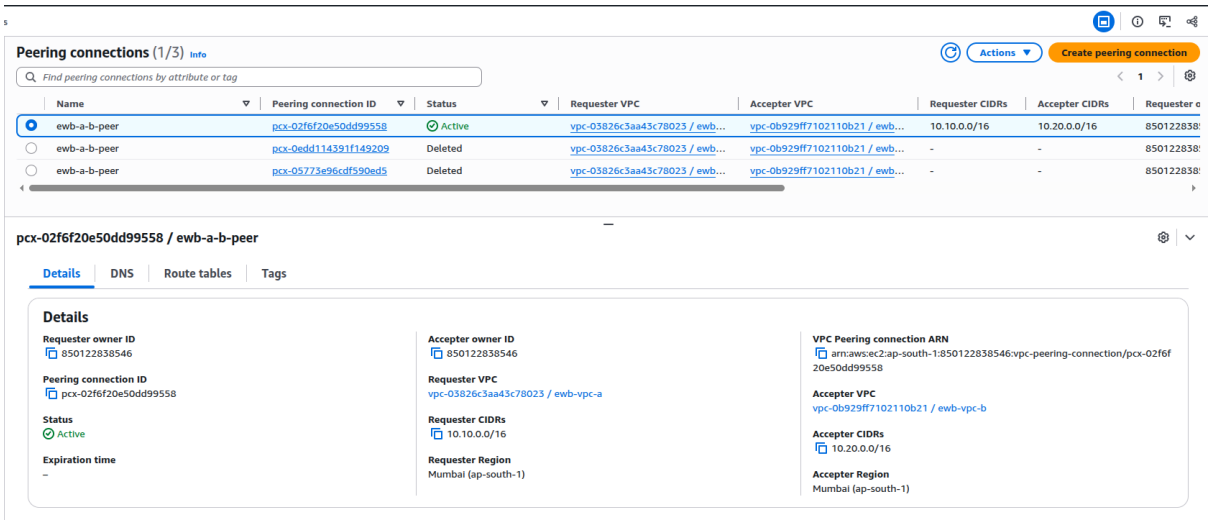


Figure 17 - Peering Connection

Routing Configuration

Route Table	Destination	Target
ewb-rt-a-private	10.10.0.0/16	local
ewb-rt-a-private	10.20.0.0/16	VPC peering connection
ewb-rt-b-private	10.20.0.0/16	local
ewb-rt-b-private	10.10.0.0/16	VPC peering connection
ewb-rt-a-public	10.10.0.0/16	local
ewb-rt-a-public	0.0.0.0/0	Internet Gateway
ewb-rt-a-public	10.20.0.0/16	VPC peering connection

The route tables were configured to provide communication between the two VPCs through the peering connection.

The private route table in VPC A contains a route for the remote network 10.20.0.0/16 targeting the VPC peering connection. The private route table in VPC B contains a return route for 10.10.0.0/16 targeting the same connection.

The public route table in VPC A contains a default route to the Internet Gateway and an additional route for 10.20.0.0/16 through the peering connection. The latter route allows the bastion host to initiate SSH connections to Application Server B through the private network path.

The local routes are retained in all route tables. Both forward and return routes are necessary because a successful communication session requires a valid path to the destination and a return path to the source.

Testing and Validation

Baseline Connectivity Test

Before establishing the VPC peering connection and configuring the peer routes, a connectivity test was performed from Application Server A to the private IP address of Application Server B.

The command **ping 10.20.1.10** was used to test reachability. The purpose of this test was to establish the initial connectivity state before implementing the required network connection.

Observed Result:

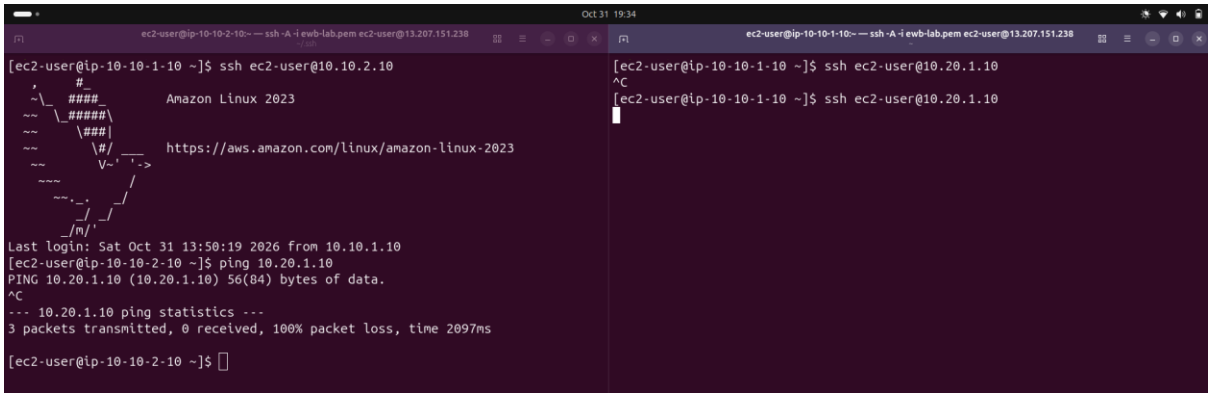


Figure 18 - Before peering condition

Forward Path Test

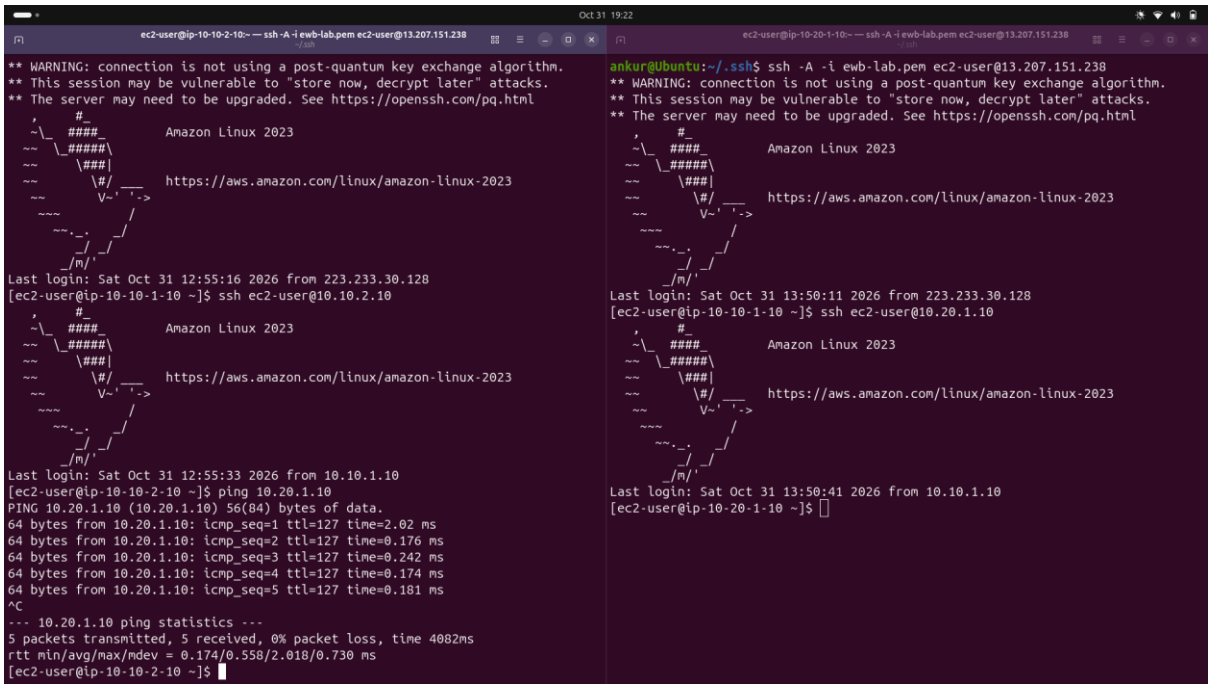


Figure 19 - Forward Path Test

Reverse Path Test

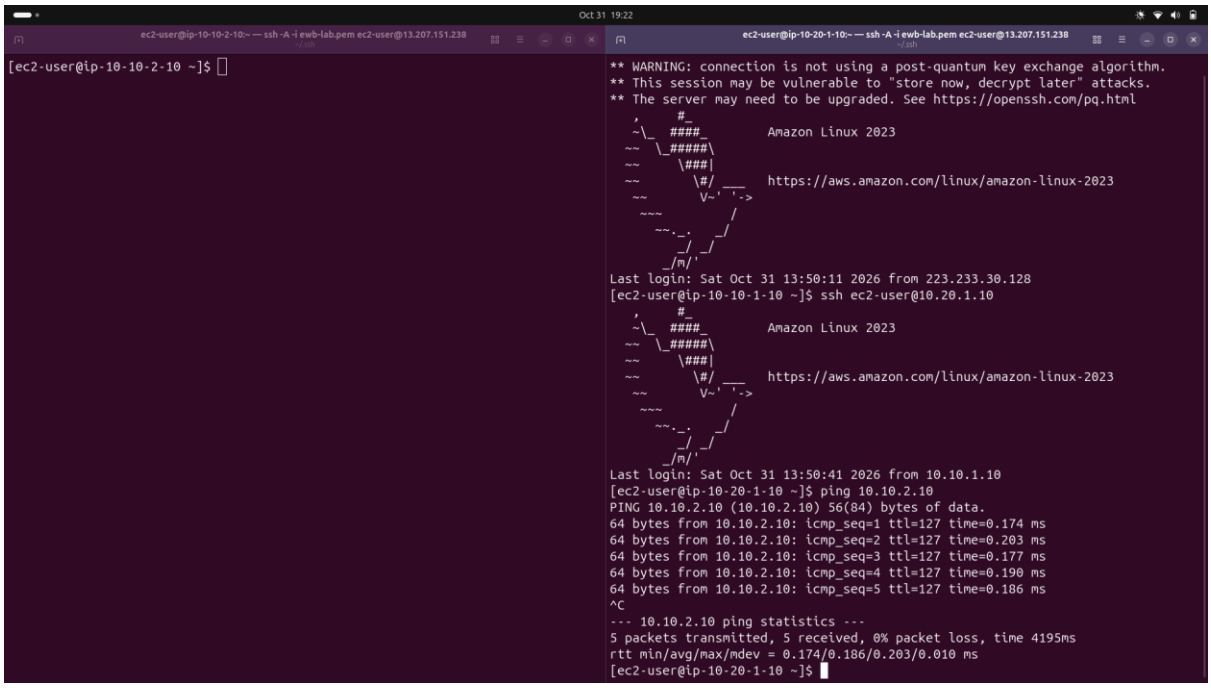


Figure 20 - Reverse Path Test

Controlled Failure and Recovery :

A controlled failure test was performed by temporarily removing the inbound ICMP Echo Request rule from the security group associated with Application Server B. The connectivity test was repeated from Application Server A to verify the effect of the security rule change.

After recording the observed result, the original ICMP rule was restored and the connectivity test was performed again.

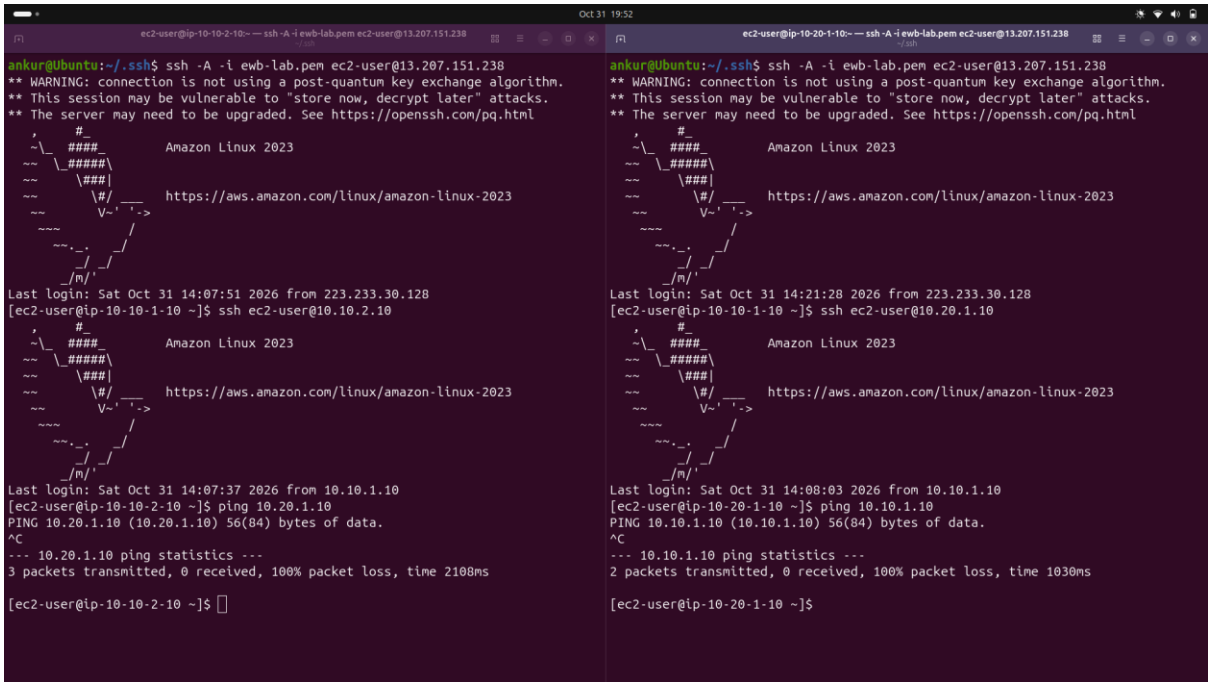


Figure 21 - After removing ICMP

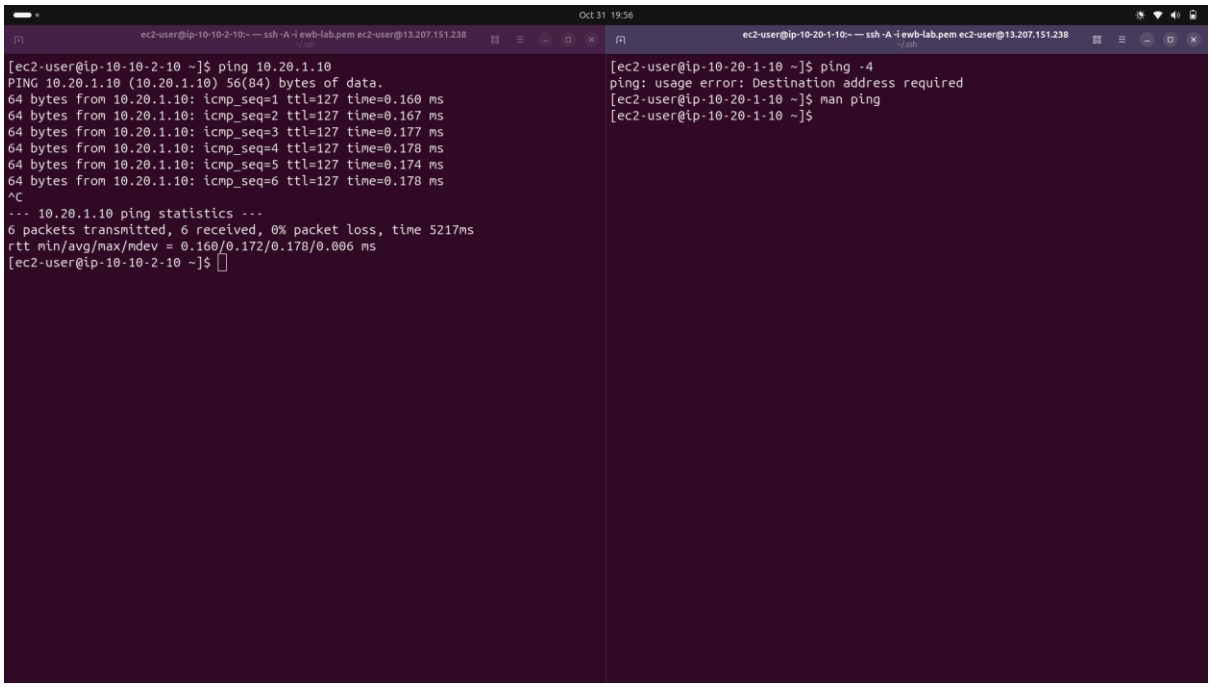


Figure 22 - After adding ICMP

Results and Discussion

Test	Expected Result (Workbook)	Actual Result
Baseline ping before peering	No replies	No replies
App-A → App-B ping	4 replies, 0% loss	5 replies, 0% loss
App-B → App-A ping	4 replies, 0% loss	5 replies, 0% loss
SSH to App-B through bastion	Successful private SSH access	Successful private SSH access
App-A/App-B public IP check	No public IPv4 on either host	No public IPv4 on either host
ICMP rule removed	Ping replies stop	Ping replies stop
ICMP rule restored	Ping replies resume	Ping replies resume

Conclusion

This project demonstrates the design and configuration of two Amazon VPCs connected through a VPC peering connection. It covers subnet configuration, Internet Gateway attachment, route table associations, security group rules, and the deployment of a public bastion host and private application servers.

The project also examines the importance of forward and return routes in enabling communication between peered VPCs. Connectivity testing before and after peering, private SSH access, and controlled security rule modification provide a practical approach to validating the network configuration.

The project highlights the importance of non-overlapping CIDR blocks, least-privilege security rules, private IP communication, and systematic troubleshooting in AWS networking.